#BuildingTheRobotsOfTheFuture

THEKER

Beyond human

Building the robots of the future

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THE PROBLEM

Robots detect objects. They still crash.

Why autonomous robots fail in warehouses, factories, and logistics hubs

Detection ≠ understanding.
They see a person. They don't know if that person is in the path or behind a barrier.

No spatial reasoning.
A forklift 20m away is not the same as one 2m ahead. Current systems treat them equally.

Real-world labels are chaos.
"hat", "helmet", "head" — three labels, same object. Messy data is the norm, not the exception.

High confidence, wrong decision.
Systems report 95% confidence and still drive into obstacles. Calibration is broken.

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THE PROBLEM

Better detectors didn't help. The gap is deeper.

WHAT'S BEEN TRIED

Scaled data 10× — detection improved. Decisions didn't.

Bigger models — more parameters, same wrong actions.

Hand-coded rules — brittle. First new scenario, they break.

WHY IT STILL FAILS

Seeing objects is not **understanding risk**

No pipeline connects **perception to decisions**

Robots need to **reason about what they see**, not just classify it

The core problem: robots can **perceive** but they can't **decide.**

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THE CHALLENGE

Industrial Robot Navigation: From Perception to Decision

"Robots fail not because they can't detect objects — they fail because they can't interpret what those objects mean, combine signals, and make robust decisions under uncertainty."

Build a complete pipeline

Perceive
Detect objects in industrial scenes. Handle 27 raw, messy label categories.

Reason
Interpret spatial context. A person far away ≠ a person in the path.

Decide
Output STOP, SLOW, or CONTINUE for every scene. Calibrate your confidence.

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WHAT YOU GET

Everything you need. Nothing you don't.

25K+ Images

17.5K train, 3.8K val, 3.8K test. Real industrial environments. COCO format.

27 Raw Categories

Messy labels by design. Grouping them is part of the challenge.

Starter Kit

Baseline pipeline included. Scores ~0.59. Beat it.

Blind Test Set

No annotations. You detect, you reason, you decide. Final ranking on hidden ground truth.

No simulation. No setup. Just build.

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EVALUATION

Can your system perceive, reason, and decide?

80%

Decision Quality

Accuracy (50%) + Confidence calibration (20%) + Reasoning quality (10%)

20%

Detection Quality

F1 score at IoU ≥ 0.5 on hidden test annotations

This is a systems challenge. The best teams will build **pipelines**, not single models.

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THE PRIZE

Win a trip to Barcelona.
Work with the team that builds real robots.

Trip to Barcelona
All expenses covered. Flights, hotel, everything.

Access to THEKER
See where real-world robotics is being built. No demo. The real thing.

Work with our engineers
Hands-on session with the team shipping perception systems.

Session with founders
Direct conversation about the future of robotics perception.

This isn't a generic prize. It's a front-row seat to real robotics.

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Robots can see.
Teach them
to think.

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